

Water Usage Inquiry: AI vs Social Media, closed loops, etc.

Posted by u/thedameisthename • 0 points • 9 comments

Sorry if this has been asked before. Searched but didn't find concrete answers to my questions. Trying to understand water usage of AI vs social media scrolling, streaming, etc.

Also, trying to find out what % of data centers use closed-loop systems vs open-loop.

If anyone has any factual responses, please provide links to sources. Thanks!

Comments

u/mrdarkbackstory • 1 points • 2026-02-18 07:33 UTC

Electrical Engineer for hyperscale datacenters here. Almost all DCs use closed loop systems. They water is recycled throughout the loop. Occasionally parts of the loop may need to be drained for maintenance, but the make up water use is negligible. Also, new technologies such as direct to chip (DTC) colling significantly reduces the amount of cooling needed.

u/EVPN • 1 points • 2026-02-17 13:01 UTC

Everything modern is closed loop. The only water a data center consumes is filling the pipes, humidification and flushing my waste.

There may be some that are closed loop with a ground source but even then it's not consuming th water. It's pulling cold ground water and putting warm water back.

The staff that operate the data center almost certainly flush more water than the data center consumes

u/Mightbeagoat2 • 3 points • 2026-02-17 02:35 UTC

<https://www.reddit.com/r/FortCollins/s/0tvSnu7HkW>

Here's a big comment I made in my local sub on my other account about this topic - kind of, but it doesn't address how much water use goes towards AI vs the rest of the internet. You could probably relate that to estimates of MW going to AI vs everything else and ratio the number below.

TL;DR is that it's hard to accurately calculate how much water DCs use. The most logical and credible estimate I could find asserts that somewhere between 800-900 Million GPD (gallons per day) goes to

all American DCs. That's a lot. But it's also a pretty small number compared to the 9 Billion GPD that Americans dump on their yards. It is inconsequentially small compared to the estimated 118 Billion GPD that goes to agriculture. Hotels, golf courses, steel production, probably microchip manufacturing & pharmaceuticals (hard to find national water use data), power generation, etc all use a lot more water than 800-900 MGD (million gallons per day)

Water is precious in a lot of places and absolutely needs to be preserved above all else. The public perception of DC water useage will be resolved as the US industry continues adopting closed loop cooling and the evaporative sites are phased out. Lots of other industries using considerably more than their fair share aren't getting any flak about it though.

Also - DCs seem to frequently get reduced to either AI or social media/memes/dumb stuff. Data centers and the cloud computing industry are literally a critical infrastructure. Everything on the internet probably lives in a DC somewhere. Banking, financial markets, industrial controls systems for non-air-gapped municipal power/water plants, hospital systems, 911 dispatch systems... some large data center outages have actually resulted in people dying. I imagine you were probably saying that harmlessly, but this is a misconception I keep seeing when people are criticizing the industry and feel obligated to correct.

u/RipEffective2538 • 5 points • 2026-02-17 01:43 UTC

They use closed loop systems now. Social media is ignorant. Closed loops use very minimal makeup. Closed loops are they way they are being built now. Especially along most of the long haul fiber lines in place

u/Dandelion-Blobfish • 1 points • 2026-02-17 01:19 UTC

You're asking great questions. This is the best analysis of data center water usage that I have come across. Most coverage on both sides is stupid for reasons explained here. I'll add that the only data centers I've come across with open loop designs were legacy facilities built 10-20 years ago in regions with abundant water supplies and climates such that a closed loop design would in net increase water "usage" when tier 2 usage (power plant) is considered.

https://youtu.be/H_c6MWk7PQc?si=AyGxi2ZQXddUMLqT

u/kubrador • 0 points • 2026-02-17 01:12 UTC

the water usage comparison is kind of apples to oranges since ai training is a one-time event and scrolling tiktok is perpetual, but yeah ai data centers slurp water like it's going out of style during training runs.

closed-loop systems are still the minority though most new builds are moving that way since water bills are no joke. the actual percentages are messy because most public disclosures are from the big hyperscalers who actually publish this stuff, and smaller operators don't.

u/Negative-Machine5718 • 0 points • 2026-02-17 00:14 UTC

From my understanding they are going to use closed loop in certain circumstances and platforms. Cost is significantly higher. There are some obvious benefits to closed loop but they don't offset cost enough for your average data center. So unless they are water constrained or in a heavily regulated environment it's likely not closed loop.

u/Lurcher99 • 3 points • 2026-02-17 04:13 UTC

All I'm building are closed loop, hyperscaler DCs.

u/danielsemaj • 2 points • 2026-02-17 00:06 UTC

I don't have any factual sources to hand other than my experience in designing building and operating (managing) hyper scale Colo data centers.

They haven't been open loop for years certainly in Europe. They use a negligible amount to pressurise the close loop and to humidify. The USA is different where the use of cooling towers is still a thing and all the big 5 hyper scalers pretty much used direct air cooling with adiabatic until recently and they still have them in operation.

They did use a lot of water for cooling up until about 6-7 years ago but it's very much changed. Indirectly they do use a lot in the power generation "scope emission" and that's pretty much the biggest amount of water they will use.